Employee Payroll and Deduction Analysis – Python Solution

Problem Statement:

Calculate an employee's gross salary using basic pay, allowances, overtime, and incentives. Apply progressive deductions based on salary slabs and generate a formatted salary statement showing gross salary, deductions, and net salary.

The solution uses only your assigned Python syllabus concepts.

1. Computational Thinking

Input:

- Employee name
- Basic salary
- Allowances
- Overtime hours and overtime rate
- Incentives

Processing:

1. Calculate total allowances.
2. Calculate overtime payment.
3. Calculate gross salary.
4. Apply progressive deductions.
5. Calculate net salary.
6. Display the salary statement.

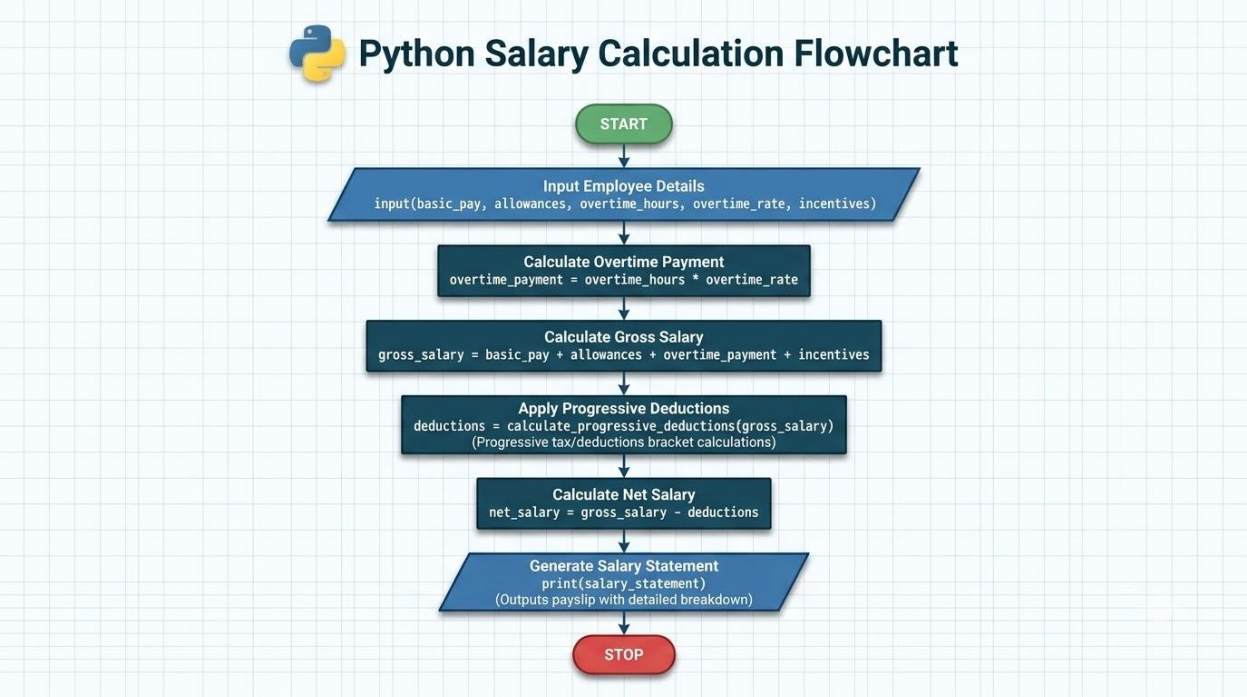
Output:

- Employee details
- Gross salary
- Deduction amount
- Net salary
- Salary classification

2. Algorithm

1. Start.
2. Read employee name.
3. Read basic salary.
4. Read allowances.
5. Read overtime hours and overtime rate.
6. Read incentives.
7. Calculate overtime payment.
8. Calculate gross salary.
9. Apply progressive deductions:
    - First ₹10,000: 0%
    - Next ₹10,000: 5%
    - Next ₹20,000: 10%
    - Above ₹40,000: 15%
10. Calculate net salary.
11. Classify the employee's salary.
12. Display the formatted salary statement.
13. Stop.

3. Flowchart



4. Pseudocode

START

INPUT employee_name

INPUT basic_salary

INPUT allowances

INPUT overtime_hours

INPUT overtime_rate

INPUT incentives

overtime_pay = overtime_hours * overtime_rate

gross_salary = basic_salary + allowances

+ overtime_pay + incentives

IF gross_salary <= 10000:

deduction = 0

ELSE IF gross_salary <= 20000:

deduction = (gross_salary - 10000) * 0.05

ELSE IF gross_salary <= 40000:

deduction = 500 + (gross_salary - 20000) * 0.10

ELSE:

deduction = 2500 + (gross_salary - 40000) * 0.15

net_salary = gross_salary - deduction

DISPLAY salary statement

STOP

5. Complete Python Program

Employee Payroll and Deduction Analysis

Input employee details

employee_name = input("Enter employee name: ")

basic_salary = float(input("Enter basic salary: "))

allowances = float(input("Enter allowances: "))

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overtime_hours = float(input("Enter overtime hours: "))
overtime_rate = float(input("Enter overtime rate per hour: "))

incentives = float(input("Enter incentives: "))

# Calculate overtime payment
overtime_pay = overtime_hours * overtime_rate

# Calculate gross salary
gross_salary = (basic_salary + allowances +
            overtime_pay + incentives)

# Progressive deduction calculation
if gross_salary <= 10000:
   deduction = 0

elif gross_salary <= 20000:
   deduction = (gross_salary - 10000) * 0.05

elif gross_salary <= 40000:
   deduction = 500 + (gross_salary - 20000) * 0.10

else:
   deduction = 2500 + (gross_salary - 40000) * 0.15

# Calculate net salary
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net_salary = gross_salary - deduction

# Salary classification
if net_salary < 15000:
    category = "Low Salary"
elif net_salary <= 30000:
    category = "Medium Salary"
else:
    category = "High Salary"

# Store salary details using a list
salary_details = [
    basic_salary,
    allowances,
    overtime_pay,
    incentives,
    gross_salary,
    deduction,
    net_salary
]

# Display formatted salary statement
print("\n====================================")
print("        EMPLOYEE SALARY SLIP")
print("====================================")
```

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print("Employee Name      :", employee_name)
print("-------------------------------------")
print("Basic Salary       : Rs.", round(salary_details[0], 2))
print("Allowances         : Rs.", round(salary_details[1], 2))
print("Overtime Payment    : Rs.", round(salary_details[2], 2))
print("Incentives         : Rs.", round(salary_details[3], 2))
print("-------------------------------------")
print("Gross Salary       : Rs.", round(salary_details[4], 2))
print("Total Deduction     : Rs.", round(salary_details[5], 2))
print("-------------------------------------")
print("Net Salary         : Rs.", round(salary_details[6], 2))
print("Salary Category     :", category)
print("=====================================")

print("      SALARY CALCULATION COMPLETED")
```

6. Sample Input

Enter employee name: Rahul

Enter basic salary: 25000

Enter allowances: 5000

Enter overtime hours: 10

Enter overtime rate: 200

Enter incentives: 2000

7. Sample Output

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====================================
        EMPLOYEE SALARY SLIP
```

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Employee Name : Rahul

Basic Salary : Rs. 25000.0

Allowances : Rs. 5000.0

Overtime Payment : Rs. 2000.0

Incentives : Rs. 2000.0

Gross Salary : Rs. 34000.0

Total Deduction : Rs. 1900.0

Net Salary : Rs. 32100.0

Salary Category : High Salary

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SALARY CALCULATION COMPLETED

8. Progressive Deduction Analysis

Gross salary slab	**Deduction rate**
First ₹10,000	0%
₹10,001–₹20,000	5%
₹20,001–₹40,000	10%
Above ₹40,000	15%

For the sample:

- Gross salary = ₹34,000
- First ₹10,000 deduction = ₹0
- Next ₹10,000 deduction = ₹500

- Remaining ₹14,000 deduction = ₹1,400
- Total deduction = ₹1,900
- Net salary = ₹32,100

9. Assigned Python Syllabus Concepts Used

Syllabus concept	**Application**
Variables	Salary and employee details
Data types	String and float
Operators	+, -, *, <=
Conditional statements	if, elif, else
Lists	Store salary details
Strings	Employee name and salary statement
Built-in functions	input(), print(), round()

Note: The deduction slabs and salary categories are illustrative rules for this programming problem, not actual income-tax rules.

Syllabus-based solution

This program avoids advanced Python concepts such as functions, classes, dictionaries, file handling, and external libraries.